NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

ANSWERS FOR THE MIDTERM EXAM

Algebra 1
2023/2024
Duration : 1h30min

Exercise 01 (6.5 points)

For each of the following statements, determine whether the statement is true or false and justify your answer.

1. If $a > 1$ is an odd positive integer, then $\log_2(a)$ is irrational.
2. For any real numbers a, b and c, if $3a + 5b \geq 11$, then $a \geq 5c + 2$ or $b \geq -3c + 1$.
3. For all integers $n \geq 10$, $n^3 < 2^n$.
4. For any two subsets $A \subset X$ and $B \subset Y$, $\complement_{X\times Y}(A \times B) = \complement_X A \times \complement_Y B$.
5. For every function $f : X \to Y$ and every subset $A \subset X$, $f(X \setminus A) = f(X) \setminus f(A)$.
6. $\sin 1°$ is irrational.

Solution

1. True. Let $a > 1$ be an odd positive integer. We have $\log_2(a) = \dfrac{\log a}{\log 2} > 0$. Assume that $\log_2(a)$ is rational. So, $\log_2(a) = \dfrac{p}{q}$ for some $(p, q) \in \mathbb{N}_0 \times \mathbb{N}$. Hence, we get $a^q = 2^p$. Since $(p, q) \in \mathbb{N}_0 \times \mathbb{N}$ and a is an odd positive integer, then a^q is an odd integer and 2^p is an even integer. Thus, $a^q = 2^p$, is a contradiction. Therefore, $\log_2(a)$ is an irrational number.
2. True. Let a, b and c be some real numbers. By the contrapositive law, the statement "if $3a + 5b \geq 11$, then $a \geq 5c+2$ or $b \geq -3c+1$" is equivalent to the statement "if $a < 5c+2$ and $b < -3c+1$, then $3a+5b < 11$". Assume that $a < 5c + 2$ and $b < -3c + 1$, then $3a + 5b < 3(5c + 2) + 5(-3c + 1) = 11$.
3. True. We have $10^3 = 1000 < 1024 = 2^{10}$. Assume that $n^3 < 2^n$. We have
$$(n + 1)^3 = n^3 \left(1 + \frac{1}{n^3}\right) < 2^n \left(1 + \frac{1}{n^3}\right) < 2^n \left(1 + \frac{1}{1^3}\right) < 2^{n+1}.$$
Thus, by principle of induction, for all integers $n \geq 10$, $n^3 < 2^n$.
4. False. Let $X := \{1, 2\}$, $Y := \{1\}$, $A := \{1\}$ and $B := \{1\}$. We have $X \times Y = \{(1, 1), (2, 1)\}$, $A \times B = \{(1, 1)\}$. So, $\complement_{X\times Y}(A \times B) = \{(2, 1)\} \neq \emptyset = \{2\} \times \emptyset = \complement_X A \times \complement_Y B$.
5. False. Let $f : \mathbb{R} \to \mathbb{R}$ be the map defined by $f(x) = x^2$ and let $A = [0, +\infty) \subset \mathbb{R}$. We have $f(\mathbb{R} \setminus [0, +\infty)) = f((-\infty, 0)) = (0, +\infty)$ but $f(\mathbb{R}) \setminus f([0, +\infty)) = \emptyset$.
6. False. Assume that $\sin 1° = \sin\left(\dfrac{\pi}{180}\right)$ is a rational number. We have $\sin\left(\dfrac{\pi}{180}\right) = \cos\left(\dfrac{\pi}{2} - \dfrac{\pi}{180}\right)$, so $\cos\left(\dfrac{89\pi}{180}\right)$ is a rational number. Then, by principle of induction, we show that $\cos\left(\dfrac{89n\pi}{180}\right)$ must be

rational for any $n \in \mathbb{N}_0$. Putting $n = 30$, we get $\cos\left(\frac{89 \times 30\pi}{180}\right) = \cos\left(14\pi + \frac{5\pi}{6}\right) = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$ is rational which is a contradiction. Hence, $\sin 1°$ is irrational.

<u>Exercise 02</u> (7.5 points)

Let f be a map defined as

$$\begin{array}{rccc} f: & \mathbb{R}^2 & \longrightarrow & \mathbb{R}^2 \\ & (x,y) & \longmapsto & \left((-1)^{\lfloor y \rfloor}\left\lfloor \frac{x}{2} \right\rfloor, x+y\right). \end{array}$$

1. Is f surjective ? Is f injective ? Justify your answers.
2. Calculate $f^{-1}(\mathbb{Z} \times \{0\})$ and $f(A \times B)$, where $A := \{x \in \mathbb{R} |\ x^2 \leq 1\}$ and $B := \{x \in \mathbb{Z} |\ |x^3 - 4x| \leq 2\}$.
3. Find two infinite subsets X and Y of $\mathbb{R}$ for which the map

$$\begin{array}{rccc} g: & X^2 & \longrightarrow & Y^2 \\ & (x,y) & \longmapsto & f(x,y) \end{array}$$

is injective.

<u>Solution</u>

1. Since for all $x \in \mathbb{R}$, $(-1)^{\lfloor y \rfloor}\left\lfloor \frac{x}{2} \right\rfloor \in \mathbb{Z}$, then for all $(x,y) \in \mathbb{R}^2$, $f(x,y) \neq (\sqrt{2}, 0)$ and so f isn't surjective. On the other hand, since $f(1,0) = (0,1) = f(0,1)$ and $(1,0) \neq (1,0)$, then f isn't injective.
2. We have

$$\begin{aligned} f^{-1}(\mathbb{Z} \times \{0\}) &= \left\{(x,y) \in \mathbb{R}^2 | f(x,y) \in \mathbb{Z} \times \{0\}\right\} \\ &= \left\{(x,y) \in \mathbb{R}^2 | (-1)^{\lfloor y \rfloor}\left\lfloor \frac{x}{2} \right\rfloor \in \mathbb{Z} \text{ and } x+y=0\right\} \\ &= \{(x,-x) | x \in \mathbb{R}\}. \end{aligned}$$

We have, $x^2 \leq 1$ is equivalent to $-1 \leq x \leq 1$ and so $A := [-1,1]$. Also, we have $x \in \mathbb{Z}$ and $|x^3 - 4x| \leq 2$ is equivalent to $x(x^2-4) \in \{-2,-1,0,1,2\}$ and x divides a, where $a \in \{-2,-1,0,1,2\}$. Hence $x(x^2-4) \in \{-2,-1,0,1,2\}$ and $x \in \{-2,-1,0,1,2\}$. Thus, we get $B = \{-2,0,2\}$. Therefore,

$$A \times B = [-1,1] \times \{-2,0,2\} = ([-1,0) \cup [0,1]) \times \{-2,0,2\} = ([-1,0) \times \{-2,0,2\}) \cup ([0,1] \times \{-2,0,2\}).$$

Thus,

$$f(A \times B) = f([-1,0) \times \{-2,0,2\}) \cup f([0,1] \times \{-2,0,2\}).$$

For $x \in [-1,0) \times \{-2,0,2\}$, we have $(-1)^{\lfloor y \rfloor}\left\lfloor \frac{x}{2} \right\rfloor = 1 \times (-1) = -1$ and $x+y =\in [-3,-2) \cup [-1,0) \cup [1,2)$, so

$$f([-1,0) \times \{-2,0,2\}) = \{-1\} \times ([-3,-2) \cup [-1,0) \cup [1,2)).$$

Similarly, for $x \in [0,1] \times \{-2,0,2\}$, we have $(-1)^{\lfloor y \rfloor}\left\lfloor \frac{x}{2} \right\rfloor = 1 \times (0) = 0$ and $x+y =\in [-2,-1] \cup [0,1] \cup [2,3]$, so

$$f([0,1] \times \{-2,0,2\}) = \{0\} \times ([-2,-1] \cup [0,1] \cup [2,3]).$$

Consequently,

$$f(A \times B) = \{-1\} \times ([-3,-2) \cup [-1,0) \cup [1,2)) \cup \{0\} \times ([-2,-1] \cup [0,1] \cup [2,3]).$$

3. Taking $(X,Y) := (2\mathbb{N},\mathbb{R})$. Then, we get

$$\begin{array}{rccc} g: & (2\mathbb{N})^2 & \longrightarrow & \mathbb{R}^2 \\ & (x,y) & \longmapsto & \left(\dfrac{x}{2}, x+y\right). \end{array}$$

Let $(x_1,y_1),(x_2,y_2) \in (2\mathbb{N})^2$ such that $g(x_1,y_1) = g(x_2,y_2)$. Thus, $\dfrac{x_1}{2} = \dfrac{x_1}{2}$ and $x_1+y_1 = x_2+y_2$. Therefore, $x_1 = x_2$ and $y_1 = y_2$. Thus, $(x_1,y_1) = (x_2,y_2)$. Hence g is injective.

Exercise 03 (6 points)

Let $\mathcal{R}$ be the binary relation defined on $\mathbb{R}$ by $x\mathcal{R}y$ if and only if $\cos^2 x + \sin^2 y = 1$.

1. Show that $\mathcal{R}$ is an equivalence relation. Is $\mathcal{R}$ a partial order relation ?
2. Determine the equivalence classes $cl\left(\dfrac{\pi}{3}\right)$ and $cl\left(\dfrac{\pi}{2}\right)$.
3. Determine $\mathbb{R}/_{\mathcal{R}}$.

Solution

1. (a) For any $x \in \mathbb{R}$, the identity $\cos^2 x + \sin^2 x = 1$ holds. Consequently, for every $x \in \mathbb{R}$, $x\mathcal{R}x$, and so $\mathcal{R}$ is reflexive.

   (b) Let $x,y \in \mathbb{R}$ such that $x\mathcal{R}y$. This implies $\cos^2 x + \sin^2 y = 1$, and so $(1-\sin^2 x)+(1-\cos^2 y) = 1$, which give us $\cos^2 y + \sin^2 x = 1$. Consequently, we get $y\mathcal{R}x$. Therefore $\mathcal{R}$ is symmetric.

   (c) Let $x,y,z \in \mathbb{R}$ such that $x\mathcal{R}y$ and $y\mathcal{R}z$. This implies $\cos^2 x + \sin^2 y = 1$ and $\cos^2 y + \sin^2 z = 1$. Consequently, combining these expressions, we obtain $\cos^2 x + \sin^2 y + \cos^2 y + \sin^2 z = 2$, leading to $\cos^2 x + \sin^2 z = 1$. As a result, we deduce that $x\mathcal{R}z$. Therefore $\mathcal{R}$ is transitive.

   Since $\mathcal{R}$ is reflexive, symmetric and transitive, then $\mathcal{R}$ is an equivalence relation. On the other hand, the identities $\cos^2 0 + \sin^2 2\pi = 1$ and $\cos^2 2\pi + \sin^2 0 = 1$ establish that $0\mathcal{R}2\pi$ and $2\pi\mathcal{R}0$. This implies that $\mathcal{R}$ isn't antisymmetric, leading to the conclusion that $\mathcal{R}$ isn't a partial order relation.

2. We have

$$\begin{aligned} cl\left(\frac{\pi}{3}\right) &= \left\{x \in \mathbb{R} \mid x\mathcal{R}\frac{\pi}{3}\right\} \\ &= \left\{x \in \mathbb{R} \mid \cos^2 x = 1 - \sin^2\left(\frac{\pi}{3}\right) = \frac{1}{4}\right\} \\ &= \left\{x \in \mathbb{R} \mid \cos x = \frac{1}{2} \text{ or } \cos x = -\frac{1}{2}\right\} \\ &= \left\{x \in \mathbb{R} \mid x = \frac{\pi}{3} + 2k\pi \text{ or } x = -\frac{\pi}{3} + 2k\pi \text{ or } x = \frac{2\pi}{3} + 2k\pi \text{ or } x = -\frac{2\pi}{3} + 2k\pi, \text{ where } k \in \mathbb{Z}\right\} \\ &= \left\{\frac{\pi}{3} + k\pi, -\frac{\pi}{3} + k\pi \mid k \in \mathbb{Z}\right\}. \end{aligned}$$

and

$$\begin{aligned} cl\left(\frac{\pi}{2}\right) &= \left\{x \in \mathbb{R} \mid x\mathcal{R}\frac{\pi}{2}\right\} \\ &= \left\{x \in \mathbb{R} \mid \cos^2 x = 1 - \sin^2\left(\frac{\pi}{2}\right) = 0\right\} \\ &= \{x \in \mathbb{R} \mid \cos x = 0\} \\ &= \left\{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\right\}. \end{aligned}$$

3. Given that, for any real number x and any integer k, $\cos^2(x + 2k\pi) = \cos^2 x$, it follows that $cl(x) = cl(x + 2k\pi)$. Additionally, using the properties $\cos^2(-x) = \cos^2 x$ and $\cos^2(x + \pi) = \cos^2(x)$, which hold for any real number x, we deduce that $cl(-x) = cl(x + \pi) = cl(x)$. Finally, for any $x, y \in \left[0, \frac{\pi}{2}\right]$ such that $x \neq y$, we have $\cos x \neq \cos y$, then $cl(x) \neq cl(y)$. Consequently, we get

$$\mathbb{R}/\mathcal{R} = \left\{cl(x) \mid x \in \left[0, \frac{\pi}{2}\right]\right\},$$

where $cl(x) = \{x + k\pi, -x + k\pi \mid k \in \mathbb{Z}\}$.